

# USE OF HYPERTONIC SODIUM CHLORIDE SOLUTIONS TO STIMULATE PERISTALSIS

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**I**N 1924, Hughson and Scarff noted the effect of intravenous sodium chloride on peristalsis. They used cats as experimental animals. As soon as a 30 per cent solution of sodium was injected intravenously, the intestine showed marked peristaltic activity. This activity persisted in some instances as long as 55 minutes. They noted the same effect with much less intensity when physiologic salt solution was used. These authors state that the use of hypertonic sodium chloride solutions injected intravenously furnish a distinctly reliable agent for promoting peristalsis in intestinal loops, which for some reason have lost their power of motor activity. In a study of the effect of several salts upon intestinal movements, Dreyer and Tsung noted that hypertonic solutions of sodium chloride caused an increase in intestinal movements. No effect was produced by the use

of isotonic sodium chloride. Carlson and Wangenstein show tracings of increased activity of the small bowel after injecting intravenously 15 cubic centimeters of 20 per cent solution of sodium chloride, but do not discuss the observation. Ross has shown both experimentally and clinically that hypertonic sodium chloride solution will produce active peristalsis. He recommends its clinical use.

In a recent series of experiments on dogs, we have compared the activity of the small intestine after giving intravenously physiologic sodium chloride solution, hypertonic sodium chloride solution, hypertonic glucose solution, and 4 per cent sodium bicarbonate solution. Our experiments were all conducted on the upper jejunum. A Thiry-Vella loop was made about 18 inches long beginning about 6 inches below the ligament of Treitz.

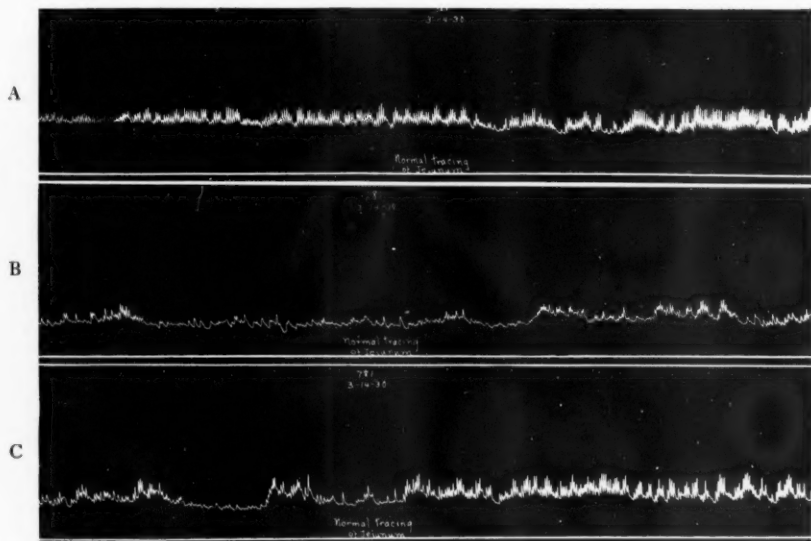


Fig. 1. Variations in the normal tracing of the jejunum. A, B, and C cover a period of one and one-half hours.

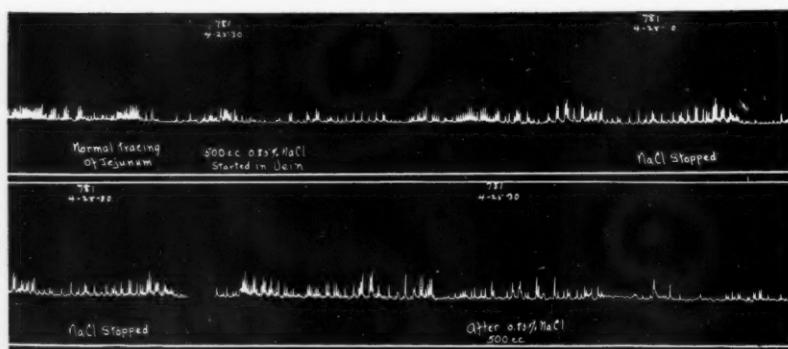


Fig. 2. Tracing of jejunum before and after intravenous administration of .85 per cent sodium chloride solution. No change is noted either in tone, rhythm, or peristalsis.

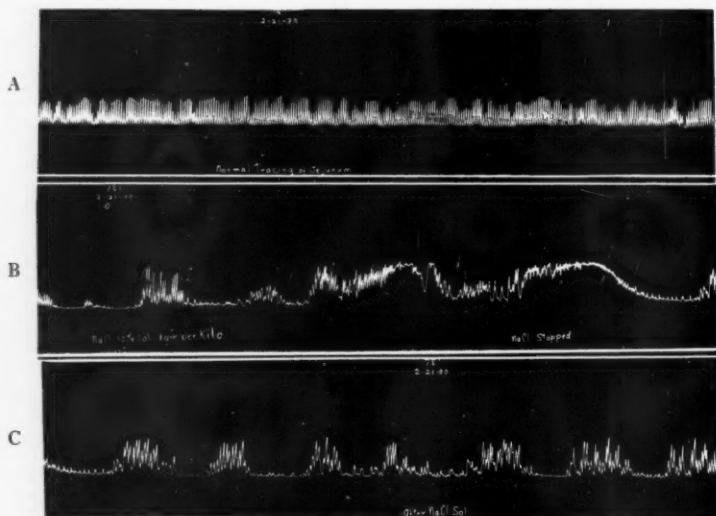


Fig. 3. Tracing of jejunum before and after intravenous injection of a 10 per cent sodium chloride solution, 1 gram per kilogram of body weight. A, Normal tracing. B, Injection of salt solution, showing increase in tone and peristalsis. C, Forty minutes after injection of hypertonic salt solution.

After complete healing had resulted following operation, tracings were begun. All tracings were made on kymograph records, a rubber bulb filled with air being used in the gut. In most cases the air pressure within the bulb and tube was kept quite uniform by measuring with a mercury column and marking on the kymograph drum before tracings were begun. Dogs were fasted for 36 hours before the

experiments so that the upper intestinal tract would be relatively empty.

#### EXPERIMENTAL RESULTS

After the intravenous administration of physiologic sodium chloride solution no change in the gut contractions was noted (Fig. 2). If hypertonic sodium chloride solution in 5 to 10 per cent strength was given,

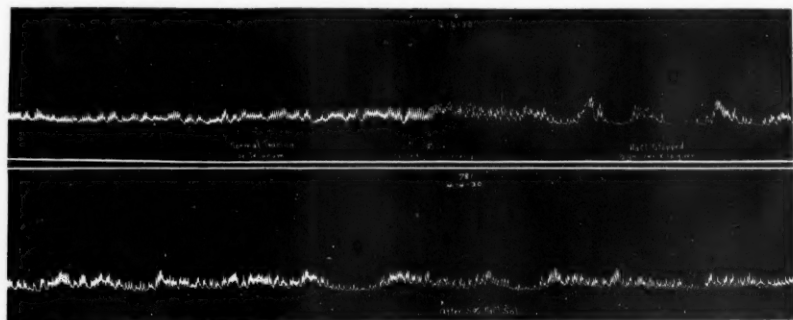


Fig. 4. Tracing of jejunum before and after intravenous administration of 5 per cent sodium chloride solution,  $\frac{1}{2}$  gram per kilogram of body weight. Slight increase in tone and peristalsis is noted.

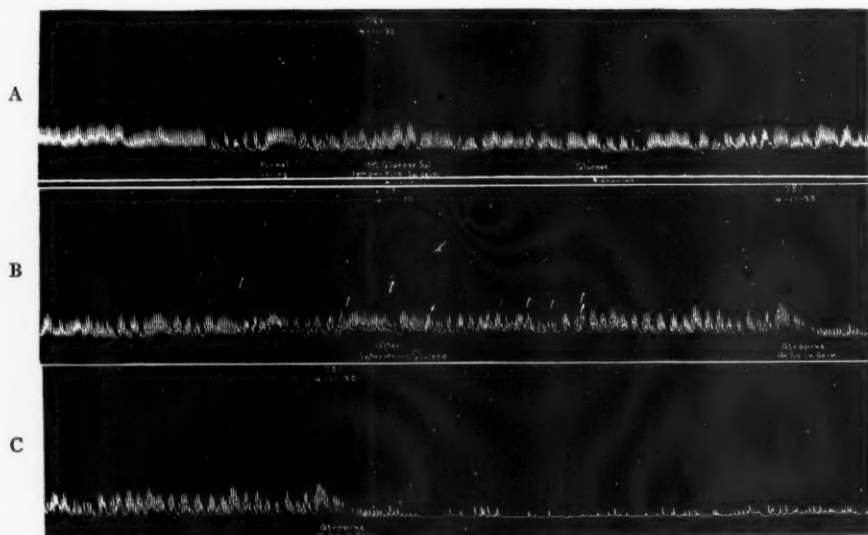


Fig. 5. Tracing of jejunum before and after intravenous injection of 10 per cent glucose solution, 1 gram per kilogram of body weight. No change is noted after glucose injection. C, Atropine, grain  $\frac{1}{50}$ , intravenously injected, causes almost immediate suspension of intestinal movement.

there always resulted an increase in the activity of the gut by raising the tone and starting active peristalsis (Figs. 3 and 4). Hypertonic glucose solution had no appreciable effect upon the gut movements (Fig. 5). A 4 per cent solution of sodium bicarbonate caused an almost immediate onset of active peristaltic waves that continued active for 30 to 45 minutes (Fig. 6). Administration of an ether anæsthetic during a jejunal tracing im-

mediately stopped the rhythmic contractions. During anæsthesia hypertonic sodium chloride solutions did not stimulate peristalsis. After the ether was removed the contractions were resumed. Atropine likewise immediately stopped contractions.

#### SUMMARY AND CONCLUSIONS

Our experiments confirm the observations of others that hypertonic sodium chloride

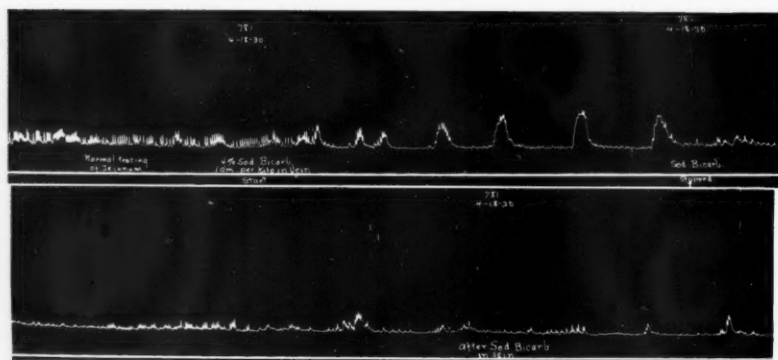


Fig. 6. Tracing of jejunum before and after intravenous administration of a 4 per cent sodium bicarbonate solution, 1 gram per kilogram of body weight. An immediate development of marked peristalsis is noted after injection of the solution. This marked change lasted from 25 to 45 minutes following injection.

solutions stimulate peristalsis. We did not find, as did Hughson and Scarff, that physiologic sodium chloride solution caused a change in the activity of the intestine. The introduction of hypertonic sodium chloride solution into the bowel caused violent peristalsis in one of our animals with vomiting and defecation. Some of our animals had bowel movements soon after the injection of hypertonic sodium chloride solution. Some appeared to have discomfort as if the increased bowel activity caused colic.

From the clinical and experimental observations it seems probable that sodium chloride introduced into the body serves to (a) replace the chlorides lost in certain diseases such as intestinal obstruction; (b) probably is an active factor in water distribution in the body; (c) stimulates peristalsis and (d), by increasing the tone of the bowel, diminishes post-operative distention and ileus with its resultant anti-peristalsis and vomiting.

A study of the action of physiologic sodium chloride solution, hypertonic sodium chloride

solution, hypertonic glucose solution and hypertonic sodium bicarbonate solution on peristalsis of the jejunum was made and the following conclusions reached; The stimulating effect of hypertonic sodium chloride solution upon peristalsis is further evidence of its value in the treatment of paralytic ileus and following operation for the relief of intestinal obstruction.

2. Further observation is necessary to determine the clinical value, if any, of sodium bicarbonate as a peristaltic stimulant.

#### BIBLIOGRAPHY

1. CARLSON, H. A., and WANGENSTEEN, O. H. Motor activity of the distal bowel in intestinal obstruction; comparison with the obstructed and normal. *Proc. Soc. Exper. Biol. & Med.*, 1930, xxvii, 676.
2. DREYER, N.B., and TSUNG, T. Effect on intestinal movements of certain salts administered intravenously. *J. Pharmacol. Exp. Therap.*, 1929, xxxvi, 629.
3. HUGHSON, W., and SCARFF, J. E. The influence of intravenous sodium chloride on intestinal absorption and peristalsis. *Johns Hopkins Hosp. Bull.*, 1924, xxxv, 197.
4. ROSS, J. W. Hypertonic saline in adynamic ileus. *Canadian M. Ass. J.*, 1926, xvi, 241.